

10. (a) Tables **A**, **B**, **C** and **D** show the results recorded by four different groups in a class investigating the reactivity of Group 7 elements (the halogens).

Each group added iodine, bromine and chlorine to solutions of sodium halides.

A tick (✓) indicates when a reaction took place and a cross (×) indicates no reaction. Only **one** table shows the expected results for this experiment.

Table **A**

|          | sodium chloride | sodium bromide | sodium iodide |
|----------|-----------------|----------------|---------------|
| bromine  | ✓               | ✓              | ×             |
| chlorine | ✓               | ×              | ✓             |
| iodine   | ✓               | ✓              | ✓             |

Table **B**

|          | sodium chloride | sodium bromide | sodium iodide |
|----------|-----------------|----------------|---------------|
| bromine  | ✓               | ✓              | ✓             |
| chlorine | ×               | ×              | ✓             |
| iodine   | ×               | ✓              | ✓             |

Table **C**

|          | sodium chloride | sodium bromide | sodium iodide |
|----------|-----------------|----------------|---------------|
| bromine  | ×               | ×              | ✓             |
| chlorine | ×               | ✓              | ✓             |
| iodine   | ×               | ×              | ×             |

Table **D**

|          | sodium chloride | sodium bromide | sodium iodide |
|----------|-----------------|----------------|---------------|
| bromine  | ×               | ✓              | ×             |
| chlorine | ✓               | ×              | ✓             |
| iodine   | ×               | ✓              | ✓             |



- (i) Give the **letter** of the table which shows the expected results for this experiment. Explain why these are the expected results. [3]

Letter .....

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- (ii) Write a balanced symbol equation for the reaction between chlorine,  $\text{Cl}_2$ , and sodium iodide solution. [3]

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- (b) A sample of iron bromide with a mass of 37.0 g contains 7.0 g of iron. Find the simplest formula of the iron bromide.

You **must** show your working. [3]

$$A_r(\text{Fe}) = 56$$

$$A_r(\text{Br}) = 80$$

Simplest formula .....

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